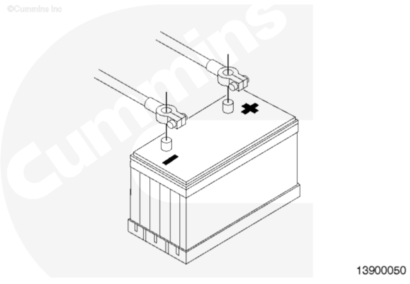
Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

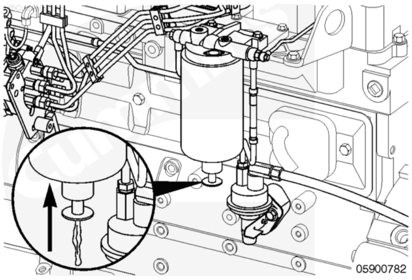


- Disconnect the batteries.

Drain

⚠ WARNING ⚠

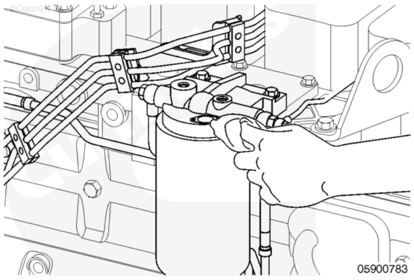
Drain the fuel-water separator into a container, and dispose of contents in accordance with local environmental regulations.



If equipped, use the filter drain valve to drain fuel out of the filter for approximately 5 seconds. This will eliminate fuel from running over the top of the filter upon removal.

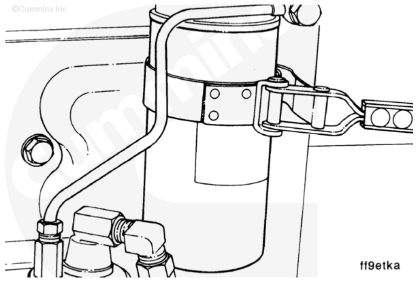
Remove

Clean the area around the fuel filter head.



⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Remove the fuel filter.

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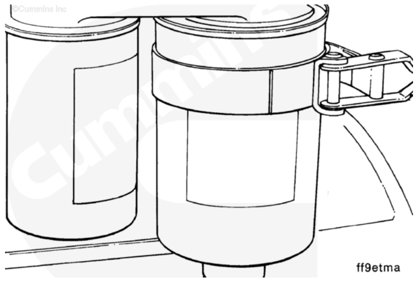
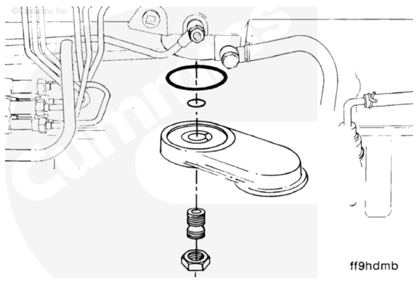


Remove the two filters from the dual-filter adapter (if equipped).

If a leak is found, replace the o-rings.

Torque Value: 32 n•m [24 ft-lb]



Install

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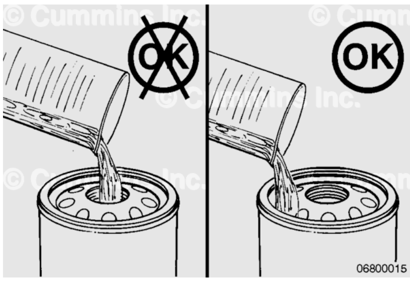
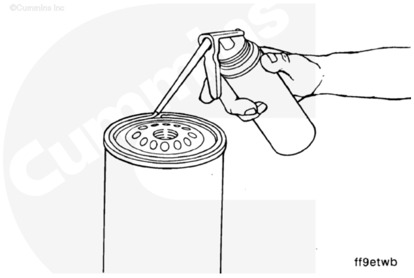


⚠ CAUTION ⚠

When pre-filling the filter do not pour fuel down the center (clean side) of the filter. Pour clean fuel into the outer openings (dirty side) of the filter. Use a clean side block off plug, if available, to prevent fuel from entering the clean side of the filter. Pre-filling on the clean side of the filter can result in debris entering the fuel system and damaging fuel system components.

Fill the new filter(s) with clean diesel fuel.

Lubricate the seal with clean lubricating engine oil.

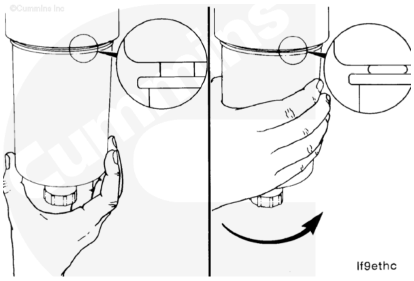


⚠ CAUTION ⚠

Mechanical overtightening can distort the threads as well as damage the filter element seal or filter can.

Install the fuel filter on the fuel filter head. Turn the filter until the gasket contacts the filter head surface.

Tighten the fuel filter an additional ½ to ¾ of a turn after the gasket contacts the fuel filter head surface, or as specified by the fuel filter manufacturer.



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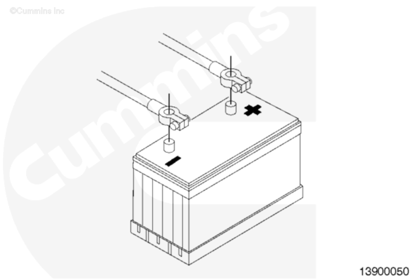
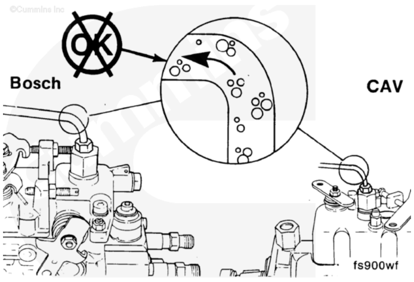


Connect the batteries.

Prime

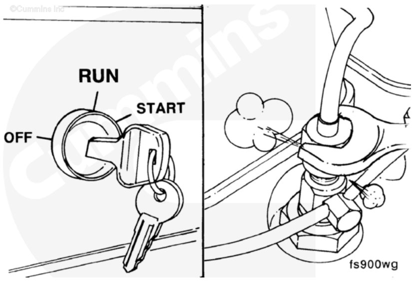
General Information

Controlled venting is provided at the injection pump through the fuel drain manifold. Small amounts of air introduced by changing the filters or injection pump supply line will be vented automatically if the fuel filter is changed in accordance with the instructions.



However, manual bleeding will be required if one of the following conditions exists:

- The fuel filter is **not** filled prior to installation
- The fuel injection pump is replaced
- The high-pressure fuel line connections are loosened, or the lines are replaced
- It is an initial engine start-up or start-up after an extended period of no engine operation.



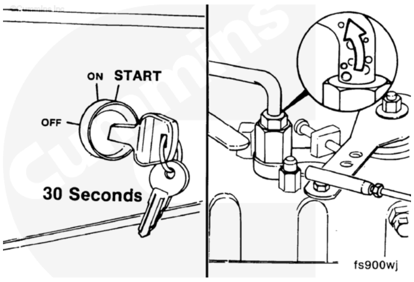
Refer to Procedure 006-015 (Fuel Filter (Spin-On)) in Section 6 for proper venting of the low pressure side of the fuel system. (/qs3/pubsys2/xml/en/procedures/40/40-006-015-tr.html)

Refer to the Procedure 005-012 (Fuel Injection Pumps, In-Line) in Section 5 to determine if venting the fuel pump is necessary. (/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html) Or, refer to Procedure 005-014 (Fuel Injection Pumps, Rotary) in Section 5 to determine if venting the fuel pump is necessary. (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html)

Refer to Procedure 006-051 Injector Supply Lines (High Pressure) in Section 6 for venting of the high pressure side of the fuel system. (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)

⚠ CAUTION ⚠

It is necessary to turn the keyswitch to the ON position. Because the engine can start, be sure to follow all safety precautions. Use the normal engine starting procedure.

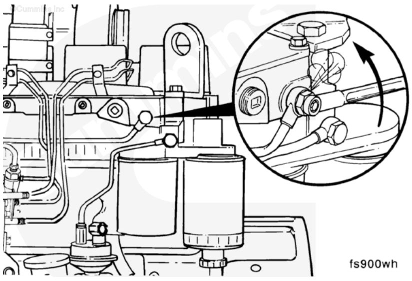


⚠ CAUTION ⚠

When using the starting motor to vent the system, do not engage the starter for more than 30 seconds, or starter damage will occur. Wait 2 minutes before starting the engine again.

Air can also be vented through the fuel drain manifold line by operating the starting motor.

Low Pressure Fuel Line(s)

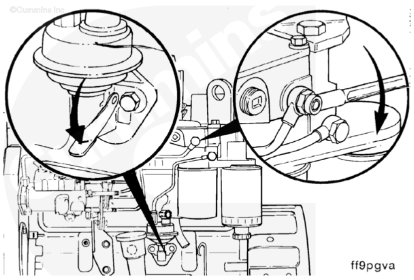


Note : For engines equipped with Distributor type pumps equipped with bleed screws.

Open the bleed screw.

Note : If the engine is not equipped with a bleed screw, loosen the fitting on the low pressure supply line to the injection pump. Once priming has been completed, tighten the fitting to the specified torque.

Operate the hand lever until the fuel flowing from the fitting is free of air.

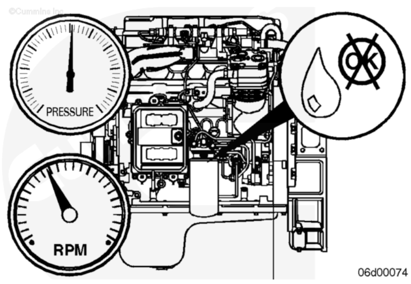


Tighten the bleed screw.

Torque Value: 9 n•m [80 in-lb]

Finishing Steps

- Operate the engine and check for leaks.



Last Modified: 16-Jan-2009